Suphia Zhang

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Education

Cornell University *Ithaca, NY*

B.S. Mechanical Engineering, Minor in Aerospace Engineering *Aug 2025 – May 2029*

- **Awards:** Meinig Family Cornell National Scholar (2025-26), McMullen Scholarship Recipient (2025-26), McMullen Dean's Scholar (2025-26), National First-Generation Recognition Program
- **Relevant Coursework:** Thermodynamics, Statics & Mechanics of Solids, Differential Equations, Heat & Mechanics

Engineering Projects & Research

NHRL Combat Robotics Team (National Havoc Robot League) *Ithaca, NY*

Mechanical Design Lead *Nov 2025 – Present*

- Led mechanical design for a 4-person team building a 3 lb beetleweight combat robot, taking it from hand sketches to a full OnShape CAD model in 6 weeks and meeting the 3 lb weight limit with 1 oz to spare.
- Iterated on the design through physical prototypes, testing each subsystem before committing to final hardware for the competition build.
- Fabricated custom machined parts and carbon fiber composite layups and assembled all subsystems to tight tolerances, enabling the robot to survive 4 competition matches without structural failure.

Cornell University Physical Intelligence (CUPI) *Ithaca, NY*

Mechanical Design Team Member *Mar 2026 – Present*

- Designed hexapod leg and tactile foot-sensing subsystems for Symbiote, a bio-inspired multi-robot system pairing an autonomous UAV quadcopter with a hexapod ground robot for coordinated terrain surveying.
- Redesigned linkages for all 6 hexapod legs and reinforced the leg carriage in Fusion 360 through iterative CAD and material analysis, increasing range of motion and load capacity to support heavier sensor payloads on unstructured terrain.
- Designed, built, and tested a Velostat taxel-array tactile foot sensor in a 3D-printed TPU housing, validating it through MuJoCo simulations and Arduino (C++)/Python scripts for real-time force visualization and terrain-adaptive feedback.

Bio-Inspired Fluid Lab *Ithaca, NY*

Undergraduate Researcher, Advisor: Dr. Sunghwan Jung *Sep 2026 – Present*

- Developing a flapping-wing aerial-aquatic vehicle modeled on MIT's FAAV, a $\sim$300 g diving-bird-inspired robot that both swims and flies, within the lab's bird design subgroup.
- Built a simulation assembly of the flapping-wing mechanism in OnShape, defining mates and joint constraints to model wing kinematics and verify range of motion before fabrication.

Work Experience

Double Dragon Restaurant *Pawtucket, RI*

Line Cook / Cashier / Staff *2015 – Aug 2026*

- Worked 60-hour weeks during academic breaks, diagnosing and repairing commercial kitchen equipment including fryers to minimize downtime during peak service.
- Onboarded 3 third-party delivery platforms (DoorDash, Grubhub, Uber Eats) in under a month, handling setup and merchant documentation, generating $\sim$200 online orders per week while managing 30+ orders per shift.

Leadership & Activities

American Society of Mechanical Engineers (ASME), Cornell University *Ithaca, NY*

Corporate Committee Member *Dec 2025 – Present*

- Conduct employer outreach and compile company dossiers to co-organize networking sessions, workshops, and speaker panels for 200+ Cornell ME students.

Skills

CAD & Design: OnShape, Fusion 360, GD&T, simulation assemblies, design iteration & validation
Fabrication: Machining, carbon fiber composite layup, 3D printing (PLA, TPU), soldering, prototyping, assembly
Programming & Software: Python, MATLAB (Onramp certified), Arduino (C++), MuJoCo, LaTeX
Languages: English (Native), Fuzhounese (Native), Mandarin (Conversational), Spanish (Conversational)